UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 001 · Introduction to partial differentiation · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·15	Deck 001, slide 15 (rendered .pptx)
T001a 00:14:20	Transcript “Introduction to Partial Differentiation”, segment start
T001b 00:11:41	Transcript “Continuity and differentiability of a function of two independent variables”
CW U1 P7	Unit 1 class-work problem 7
ESA Dec-23 1a	Past paper: Dec 2023, question 1.a

Contents

1. Figure G1.1 · Graph of $z = \sin x \cos y$
2. Figure G1.2 · Graph of $z = e^{1-x^2-y^2}$
3. Figure G2.1 · Paths along which (x, y) can approach (a, b)
4. Figure G2.2 · $\frac{2xy}{3x^2+y^2}$ along the line $y = mx$
5. Figure G3.1 · $z = \frac{3x^2y}{x^2+y^2}$: the limit at the origin exists
6. Figure G3.2 · $z = \frac{2xy}{3x^2+y^2}$: the limit at the origin does not exist

Colours follow the Okabe–Ito order: blue = main surface or curve, vermilion = comparison, green and purple = further paths. Figures G1.1–G1.2 and G3.1–G3.2 redraw the deck's Gnuplot surfaces ($\sin x \cos y$, $e^{1-x^2-y^2}$, $\frac{3x^2y}{x^2+y^2}$, $\frac{2xy}{3x^2+y^2}$) over $-2 \le x, y \le 2$ or a similar range; the slides' screenshots are not copied. Figure G2.1 redraws the lecture's board sketch of paths approaching (a, b).

1 Functions of two or more variables

$z = \sin x \cos y$

$-\pi \le x,\ y \le \pi;\ \ -1 \le z \le 1$

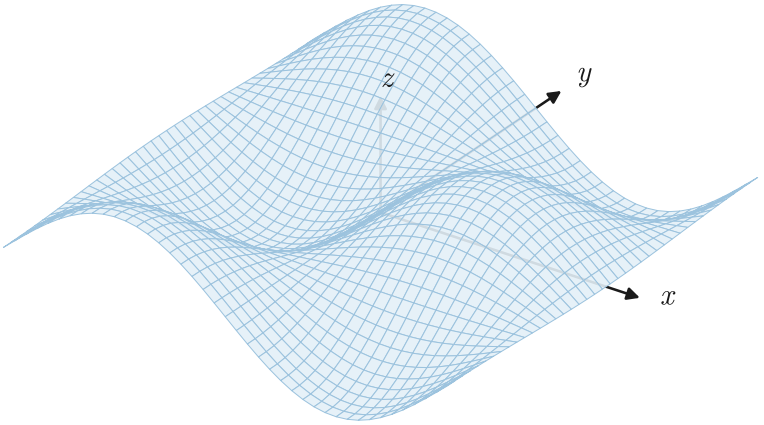


Figure G1.1. Graph of $z = \sin x \cos y$. ESA ★

S001·8 · T001a 00:09:07

How to read it. A function of two variables is a surface over the xy-plane: above each point (x, y) the height is $f(x, y)$. $\sin x \cos y$ is an egg-box of hills and hollows between -1 and 1; the slide's Gnuplot picture shows the same surface.

$z = e^{1 - x^2 - y^2}$

top $(0, 0, e)$

green: level curves $z =$ const (circles)

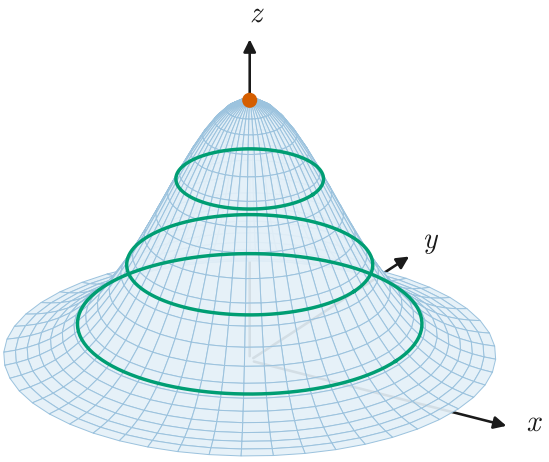


Figure G1.2. Graph of $z = e^{1-x^2-y^2}$. ESA ★

S001·8 · T001a 00:09:07

How to read it. A single bell-shaped hill with its top $z = e$ above the origin, falling to 0 in every direction. The green level curves (drawn on the surface) are circles $x^2 + y^2 =$ constant.

2 Limit of a function of two variables

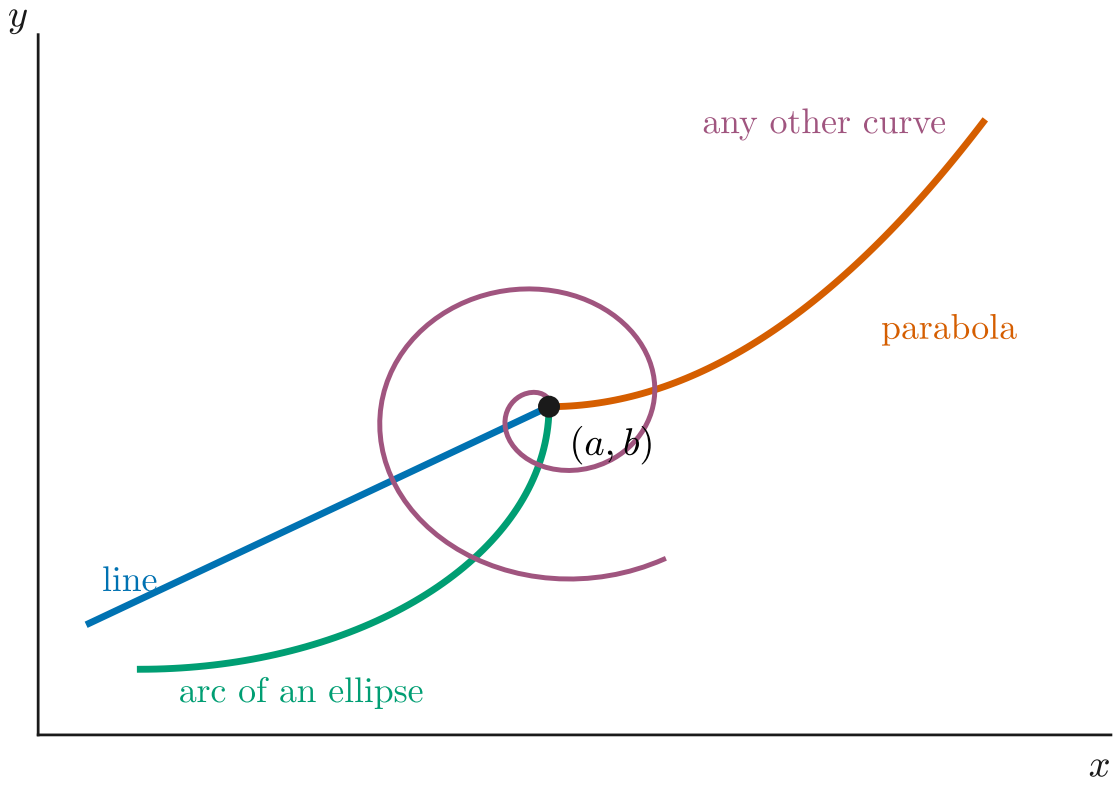


Figure G2.1. Paths along which (x, y) can approach (a, b). ESA ★

S001·9 · T001a 00:11:52 · T001a 00:12:19

How to read it. In one variable x reaches a only from the left or the right. In the plane, (x, y) can reach (a, b) along a line, a parabola, an arc of an ellipse, or any other path. The limit exists only if all of them give the same value.

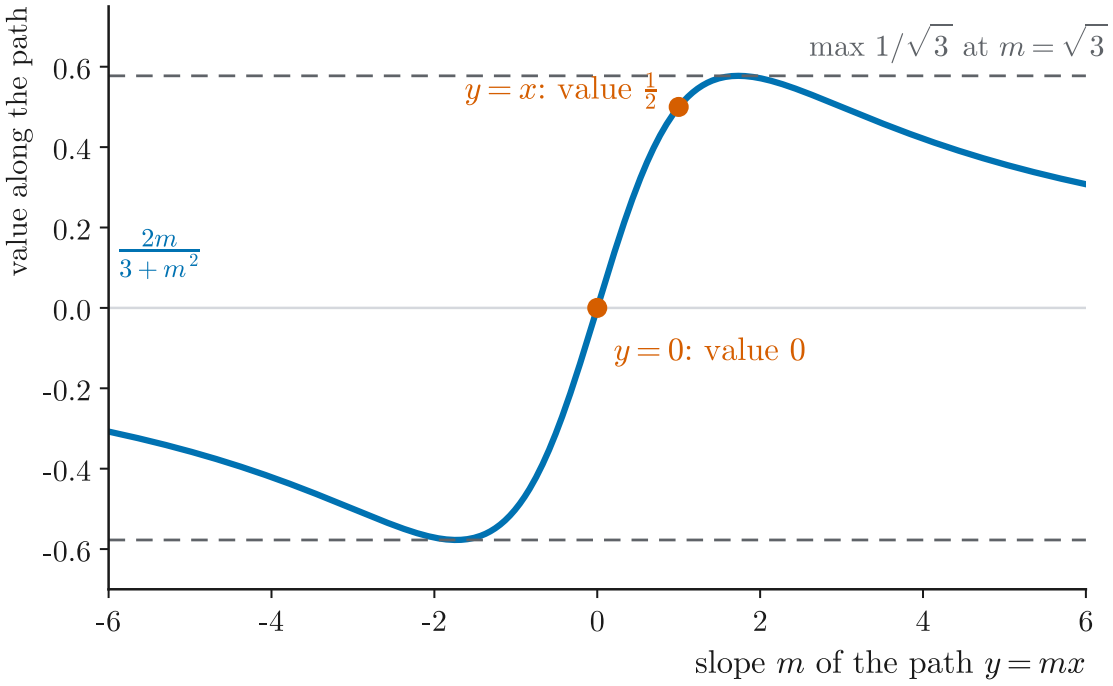


Figure G2.2. $\frac{2xy}{3x^2+y^2}$ along the line $y=mx$. ESA ★★

ADDED FOR CLARITY S001·10 · T001a 00:17:24

How to read it. The value of $\frac{2xy}{3x^2+y^2}$ on the line $y=mx$ is the constant $\frac{2m}{3+m^2}$, plotted against the slope m. $m=0$ gives 0, $m=1$ gives $\frac{1}{2}$, and the largest value is $\frac{1}{\sqrt{3}}$ at $m=\sqrt{3}$. Because the value changes with the path, the limit at the origin does not exist.

3 Continuity of a function of two variables

$$z = \frac{3x^2 y}{x^2 + y^2}$$

limit 0 at the origin; $f(0,0)$ undefined

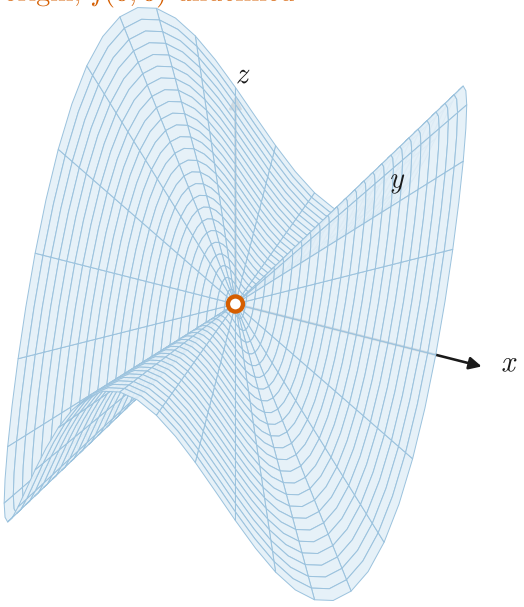


Figure G3.1. $z = \frac{3x^2y}{x^2+y^2}$: the limit at the origin exists. ESA ★

S001·12 · T001b 00:02:41

How to read it. The surface folds near the origin but every direction leads to height 0 there (vermilion point). It is missing only the single point $(0,0)$, where f is undefined: a removable discontinuity.

$z = \dfrac{2xy}{3x^2 + y^2}$

ray $y = x$: height $\frac{1}{2}$

x-axis: height 0

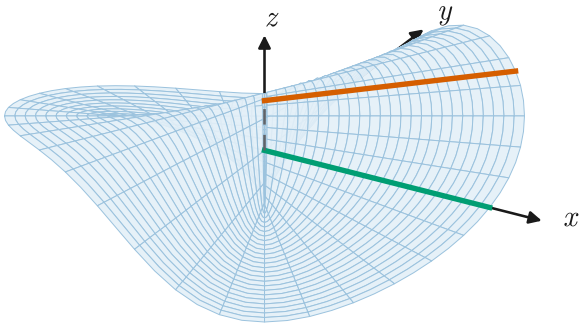


Figure G3.2. $z = \frac{2xy}{3x^2+y^2}$: the limit at the origin does not exist. ESA ★

S001·13 · T001b 00:06:13

How to read it. Each ray from the origin keeps a constant height $\frac{2m}{3+m^2}$, so the surface is a twisted fan of straight lines meeting at the z-axis at different heights (vermilion: the ray $y = x$ at height $\frac{1}{2}$; blue: the x-axis at height 0). There is no single value at the origin.